Azure Key Vault Keys – Encrypting Azure Storage Account Using Customer-Managed Key

1. What We Will Learn

In this demo, we will understand:

1. What is a Key in Azure Key Vault?
2. What are cryptographic operations?
3. Difference between Microsoft-managed keys and customer-managed keys.
4. How to use an Azure Key Vault key to encrypt a Storage Account.
5. Why the Storage Account needs permission to access the Key Vault key.
6. How **Managed Identity** is used to provide access.
7. How to assign the **Key Vault Crypto Officer** role.
8. How the Storage Account becomes encrypted using a customer-managed key.

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2. What is a Key in Azure Key Vault?

A **Key** in Azure Key Vault is used for **cryptographic operations**.

In simple terms, a key can be used for:

- Encryption
- Decryption
- Digital signing
- Verification

For this demo, we are mainly focusing on **encryption**.

Simple Example

```
Data
  |
  | Encryption Key
  ↓
Encrypted Data
```

The encryption key can be securely managed in Azure Key Vault.

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3. Why Do We Need Azure Key Vault for Encryption?

Suppose we have an Azure Storage Account.

We want to encrypt the data stored in that Storage Account.

```
Azure Storage Account
        |
        | Encryption
        ↓
      Data
```

For encryption, we need an encryption key.

Instead of managing the key directly inside the Storage Account, we can store and manage the key in **Azure Key Vault**.

```
Azure Storage Account
        |
        | Uses Key
        ↓
 Azure Key Vault
        |
        ↓
 Encryption Key
```

This allows us to centrally manage and control the cryptographic key.

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4. Important Concept – Storage Account Needs Permission

There is an important point here.

The Storage Account needs permission to access and use the key stored in Azure Key Vault.

Without permission:

```
Storage Account
       |
       | Request Key
       ↓
 Azure Key Vault
       |
       X
Access Denied
```

With the required permission:

```
Storage Account
       |
       | Authorized Access
       ↓
 Azure Key Vault
       |
       ↓
 Encryption Key
```

Therefore, we need to provide the required permission to the Storage Account.

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5. What is Managed Identity?

Managed Identity is an Azure-provided identity that allows an Azure resource to authenticate to other Azure services without us having to manage credentials such as usernames, passwords, or client secrets.

In this example:

```
Storage Account
       |
       | Managed Identity
       ↓
 Azure Key Vault
```

The Storage Account uses its managed identity to access the Key Vault.

Simple Understanding

Think of Managed Identity as:

> "An identity automatically provided to an Azure resource so that the resource can securely access another Azure service."

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6. Microsoft-Managed Keys vs Customer-Managed Keys

When creating a Storage Account, Azure provides encryption options.

Two important concepts are:

Microsoft-Managed Keys

Azure manages the encryption keys for us.

```
Storage Account
       |
       ↓
Microsoft-Managed Key
```

We don't have to manage the encryption key ourselves.

Customer-Managed Keys

We manage the encryption key using a supported key-management service such as Azure Key Vault.

```
Storage Account
       |
       | Uses Key
       ↓
 Azure Key Vault
       |
       ↓
Customer-Managed Key
```

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7. Demo – Encrypt Storage Account Using Key Vault Key

Step 1: Create a Storage Account

Go to Azure Portal and create a Storage Account.

Enter the required details such as:

- Subscription
- Resource Group
- Storage Account Name
- Region
- Performance
- Redundancy

During creation, go to:

```
Data Protection / Encryption
```

For the initial setup, select:

Microsoft-managed keys

Then create the Storage Account.

---

8. Step 2: Create a Key in Azure Key Vault

We already have an Azure Key Vault.

Inside the Key Vault:

```
Key Vault
    ↓
Objects
    ↓
Keys
```

We have a key such as:

```
StorageAccountEncryptionKey
```

This key will be used to encrypt the Storage Account.

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9. Step 3: Open Storage Account Encryption Settings

Open the Storage Account.

Go to:

```
Security + Networking
       ↓
Encryption
```

Initially, the encryption type will be:

Microsoft-managed keys

We want to change this to:

Customer-managed keys

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10. Step 4: Select Customer-Managed Keys

Select:

Customer-managed keys

Now Azure asks us to select:

1. Key Vault
2. Key

For example:

```
Key Vault:
OurFirstKeyVault
```

```
Key:
StorageAccountEncryptionKey
```

Then click:

Select

and try to save the configuration.

---

11. Why Does the Save Operation Fail?

We may receive an error because the Storage Account does not have the required permission to access the Key Vault key.

Remember:

```
Storage Account
       |
       | Needs permission
       ↓
 Azure Key Vault
       |
       ↓
 Encryption Key
```

The Storage Account must be authorized before it can use the key.

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12. Step 5: Assign Permission to Storage Account

Go to the Azure Key Vault.

Then:

```
Key Vault
   ↓
Access control (IAM)
   ↓
Add
```

```
    ↓
Add role assignment
```

Search for:

```
Key Vault Crypto Officer
```

Select the role.

---

13. Why Key Vault Crypto Officer?

The **Key Vault Crypto Officer** role provides permissions for cryptographic key operations according to the role definition.

In simple terms:

```
Key Vault Crypto Officer
          |
          ↓
       Key Access
          |
          ↓
Cryptographic Operations
```

For this demo, we assign this role to the Storage Account's managed identity.

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14. Assign the Role to Managed Identity

After selecting:

Key Vault Crypto Officer

Click:

```
Next
```

For the member type, select:

Managed identity

Then select:

Members

From the resource type/dropdown, select:

Storage Account

Then select our Storage Account.

For example:

```
Managed Identity
       |
       ↓
Storage Account
```

Then:

```
Review + Assign
       ↓
Assign
```

The role assignment is now created.

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15. Why Are We Assigning the Role to Managed Identity?

This is one of the most important concepts in this demo.

We are **not simply giving permission to a human user**.

We are giving permission to the **Storage Account's managed identity**.

The flow becomes:

```
Storage Account
       |
       | Managed Identity
       ↓
```

```
Azure Key Vault
       |
       | Permission
       ↓
Storage Account Encryption Key
```

Therefore, the Storage Account can access the required key.

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16. Important – Role Assignment May Take Time

After assigning the role, the permission may not become effective immediately.

It can take some time for the role assignment to propagate.

Therefore, wait for a few minutes and then try the operation again.

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17. Step 6: Configure Customer-Managed Key Again

Go back to:

```
Storage Account
        ↓
Security + Networking
        ↓
Encryption
```

Select:

Customer-managed keys

Then select:

```
Key Vault
        ↓
Our Key Vault
        ↓
StorageAccountEncryptionKey
```

Click:

Select

Then click:

Save

---

18. Result

This time, the operation should succeed.

The Storage Account is now configured to use the selected customer-managed key.

The overall flow is:

```
                     Azure Key Vault
                            |
                            |
                Storage Account Key
                            |
                            ↓
                Customer-Managed Key
                            |
                            |
                   Azure Storage
                            |
                            ↓
                        Data
```

---

19. Complete Architecture

```
                   Azure Storage Account
                            |
                            |
                    Managed Identity
                            |
                            ↓
                    Azure Key Vault
                            |
```

```
                |
   Storage Encryption Key
                |
                ↓
          Encryption
                |
                ↓
         Storage Data
```

---

20. Microsoft-Managed vs Customer-Managed Keys

Microsoft-Managed Keys	Customer-Managed Keys
Microsoft/Azure manages the keys	Customer controls the key
Less management required	More control and management required
Simple to configure	Requires additional configuration
No need to manage Key Vault key permissions for this purpose	Requires appropriate Key Vault access
Suitable for many standard scenarios	Useful when organizations need greater control over encryption keys

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21. Important Interview Question

Q1. Why does a Storage Account need permission to access a Key Vault key?

Because the encryption key is stored in Azure Key Vault. The Storage Account must be authorized to use that key for encryption-related operations.

Q2. How can we provide permission to the Storage Account?

We can use the Storage Account's **Managed Identity** and assign the appropriate Azure RBAC role on the Key Vault.

Q3. Why do we use Managed Identity?

Managed Identity allows an Azure resource to authenticate to another Azure service without us having to manually manage credentials such as passwords or client secrets.

Q4. What is a Customer-Managed Key?

A Customer-Managed Key is an encryption key that the customer controls and manages, commonly using Azure Key Vault.

Q5. What is the difference between Microsoft-Managed and Customer-Managed Keys?

Microsoft-managed key:

Azure manages the encryption key for us.

Customer-managed key:

The customer manages the encryption key and controls access to it.

Q6. Where is the Customer-Managed Key stored?

In this scenario, the encryption key is stored and managed in **Azure Key Vault**.

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22. Most Important Flow to Remember

Remember this flow for interviews:

```
1. Create Storage Account
          ↓
2. Create Key in Key Vault
          ↓
3. Select Customer-Managed Key
          ↓
4. Select Key Vault + Key
          ↓
5. Operation fails if Storage lacks permission
```

```
          ↓
6. Enable/Use Managed Identity
          ↓
7. Assign required Key Vault RBAC role
          ↓
8. Wait for role propagation
          ↓
9. Select Customer-Managed Key again
          ↓
10. Save
          ↓
11. Storage Account uses the Key Vault key
```

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23. Quick Revision

Key

Key = Used for cryptographic operations such as encryption and decryption.

Managed Identity

Managed Identity = Azure-managed identity that allows an Azure resource to securely authenticate to other Azure services without managing credentials manually.

Customer-Managed Key

Customer-Managed Key = Encryption key controlled and managed by the customer, commonly through Azure Key Vault.

Key Vault Crypto Officer

Key Vault Crypto Officer = An Azure RBAC role that provides permissions for cryptographic key operations according to the role definition.

Final Architecture

```
Storage Account
      |
      | Managed Identity
      ↓
Azure Key Vault
      |
      | Customer-Managed Encryption Key
```

```
       ↓
Storage Account Data
       |
       ↓
    Encrypted
```

One-Line Interview Answer

> "We can encrypt an Azure Storage Account using a customer-managed key stored in Azure Key Vault. The Storage Account uses its managed identity to authenticate to Key Vault, and we assign the required RBAC permissions so that the Storage Account can use the encryption key."